RPS6KA4 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes RPS6KA4 gene fusions from the msk_impact_50k_2026 study, covering 13 fusion events. 5/8 fusions with known frame status (62.5%) were in-frame. 7/12 fusions (58.3%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.261905, not statistically significant at alpha=0.05). No literature benchmark is configured for RPS6KA4.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 13 analyzed fusion events, identifying 12 distinct fusion partner genes. The most frequent partner was BAD, observed in 2 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 4/5 (80.0%) of in-frame fusions retained the domain, p=0.261905 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.415842.

Domain disruption

Disruption of the Protein kinase domain and Protein kinase C terminal domain was tested with Fisher's exact test comparing in-frame fusions against all others; 1/4 (25.0%) of in-frame fusions disrupted the domain, p=0.97619 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.970297.

Junction concentration

50% of 6 fusions with this gene as the 3' partner join at exon 6, against 2% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.003996, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 10 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 209 aa (exon 6; label-permutation p=0.0719281). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -92 aa from the nearest configured domain boundary at 301 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=5) and "not_retained" (n=5). For Frame_status, retained: 3/4 (75.0%, 95% CI 30.1%-95.4%); not_retained: 1/3 (33.3%, 95% CI 6.1%-79.2%). Welch's t-test comparing Tumor_variant_count between groups gave means of 30.8 vs 51.4, p=0.362096 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for RPS6KA4 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (6 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: RPS6KA4 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: RPS6KA4 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (13 row(s) in window_scan, 5 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
BAD	2	0.15384615384615385	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.10201363134822938	1
DOCK2	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	2
EEFSEC	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	3
FERMT3	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	4
GPR137	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	5
MACROD1	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	6
MARK2	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	7
PLEKHA7	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	8
RPLP0P2	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	9
SLC22A9	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	10
STIP1	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TTYH2	1	0.07692307692307693	0.03810720833847421	0.001309529809014773	None	0.30769032445727307	confidence_certainty, domain_disruption, domain_retention, recurrence	0.07486431008126106	12

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
19	1	4	0	5	None	1.0	-0.0
191	3	2	0	5	None	0.16666666666666666	0.7781512503836436
209	4	1	0	5	None	0.04761904761904762	1.3222192947339193
217	4	1	1	4	16.0	0.2063492063492064	0.6853971971467449
358	5	0	1	4	None	0.047619047619047616	1.3222192947339193
589	5	0	2	3	None	0.16666666666666666	0.7781512503836436
599	5	0	3	2	None	0.4444444444444445	0.3521825181113625
653	5	0	4	1	None	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

1	4
3	2

frame_domain_contingency_tables_by_tumor_type

stratum	table
IDC	[1, 0], [2, 0]
LUAD	[0, 1], [0, 0]
LUCA	[0, 0], [0, 1]
MFH	[0, 1], [0, 0]
PAAD	[0, 0], [1, 0]
READ	[0, 0], [0, 1]
STAD	[0, 1], [0, 0]
TCCA	[0, 1], [0, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf00069	Protein kinase domain	10	5	4	1	0.5868217054263566
auto_disruption_pf00433	Protein kinase C terminal domain	10	7	1	2	0.05925925925925926

domain_retention tables

frame_domain_contingency_table

3	2
1	4

frame_domain_contingency_tables_by_tumor_type

stratum	table
IDC	[2, 0], [1, 0]
LUAD	[0, 0], [0, 1]
LUCA	[0, 1], [0, 0]
MFH	[0, 0], [0, 1]
PAAD	[1, 0], [0, 0]
READ	[0, 1], [0, 0]
STAD	[0, 0], [0, 1]
TCCA	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	10	5	4	1	0.5868217054263566

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	8	11	0.7272727272727273

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
BAD	2	2	2	1
DOCK2	1	1	1	1
EEFSEC	1	1	1	1
FERMT3	1	1	1	1
GPR137	1	1	1	1
MACROD1	1	1	1	1
MARK2	1	1	1	1
PLEKHA7	1	1	1	1
RPLP0P2	1	1	1	1
SLC22A9	1	1	1	1
STIP1	1	1	1	1
TTYH2	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
11	64128000	64129000	3	EVT-P-0007565-T02-IM6-1, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6
11	64137000	64138000	3	EVT-P-0022310-T01-IM6-5, EVT-P-0048702-T01-IM6-21, EVT-P-0050067-T01-IM6-20
11	64126000	64127000	1	EVT-P-0031756-T01-IM6-4
11	64129000	64130000	1	EVT-P-0040997-T01-IM6-11
11	64135000	64136000	1	EVT-P-0012674-T01-IM5-10
11	64138000	64139000	1	EVT-P-0027460-T01-IM6-12

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
191	2	2	2	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	6	3	6	0.5	0.023610578765810655	3.602899932444442
five_prime	6	1	4	0.25	0.01326178612495209	1.283986689375516
five_prime	14	1	4	0.25	0.0403219624377156	0.8187503329098432
five_prime	15	1	4	0.25	0.025910310463779226	1.0013819283183554
five_prime	16	1	4	0.25	0.055193560751245686	0.6921627813121456
three_prime	2	1	6	0.16666666666666666	0.012188577999233422	1.149115353615786
three_prime	10	1	6	0.16666666666666666	0.21425833652740514	0.11652637789250317
three_prime	14	1	6	0.16666666666666666	0.02100421617477961	0.9223048542708715

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
9	209	200	4	6	4	1	None	0.04761904761904762	1.3222192947339193	EVT-P-0007565-T02-IM6-1, EVT-P-0031756-T01-IM6-4, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6
17	217	200	5	5	4	1	16.0	0.2063492063492064	0.6853971971467449	EVT-P-0007565-T02-IM6-1, EVT-P-0031756-T01-IM6-4, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11
19	219	200	5	5	4	1	16.0	0.2063492063492064	0.6853971971467449	EVT-P-0007565-T02-IM6-1, EVT-P-0031756-T01-IM6-4, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11
117	217	100	4	6	3	2	6.0	0.5238095238095238	0.2808266095756942	EVT-P-0007565-T02-IM6-1, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11
158	358	200	5	5	4	1	16.0	0.2063492063492064	0.6853971971467449	EVT-P-0007565-T02-IM6-1, EVT-P-0012674-T01-IM5-10, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11
167	217	50	4	6	3	2	6.0	0.5238095238095238	0.2808266095756942	EVT-P-0007565-T02-IM6-1, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11
191	241	50	4	6	3	2	6.0	0.5238095238095238	0.2808266095756942	EVT-P-0007565-T02-IM6-1, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
191	291	100	4	6	3	2	6.0	0.5238095238095238	0.2808266095756942	EVT-P-0007565-T02-IM6-1, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11
191	391	200	5	5	4	1	16.0	0.2063492063492064	0.6853971971467449	EVT-P-0007565-T02-IM6-1, EVT-P-0012674-T01-IM5-10, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11
465	665	200	4	6	0	5	0.0	0.04761904761904762	1.3222192947339193	EVT-P-0022310-T01-IM6-5, EVT-P-0027460-T01-IM6-12, EVT-P-0048702-T01-IM6-21, EVT-P-0050067-T01-IM6-20
565	665	100	4	6	0	5	0.0	0.04761904761904762	1.3222192947339193	EVT-P-0022310-T01-IM6-5, EVT-P-0027460-T01-IM6-12, EVT-P-0048702-T01-IM6-21, EVT-P-0050067-T01-IM6-20
589	689	100	4	6	0	5	0.0	0.04761904761904762	1.3222192947339193	EVT-P-0022310-T01-IM6-5, EVT-P-0027460-T01-IM6-12, EVT-P-0048702-T01-IM6-21, EVT-P-0050067-T01-IM6-20
589	789	200	4	6	0	5	0.0	0.04761904761904762	1.3222192947339193	EVT-P-0022310-T01-IM6-5, EVT-P-0027460-T01-IM6-12, EVT-P-0048702-T01-IM6-21, EVT-P-0050067-T01-IM6-20

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
565	665	100	4	6	0	5	0.0	0.04761904761904762	1.3222192947339193	EVT-P-0022310-T01-IM6-5, EVT-P-0027460-T01-IM6-12, EVT-P-0048702-T01-IM6-21, EVT-P-0050067-T01-IM6-20
9	209	200	4	6	4	1	None	0.04761904761904762	1.3222192947339193	EVT-P-0007565-T02-IM6-1, EVT-P-0031756-T01-IM6-4, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6
17	217	200	5	5	4	1	16.0	0.2063492063492064	0.6853971971467449	EVT-P-0007565-T02-IM6-1, EVT-P-0031756-T01-IM6-4, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11
158	358	200	5	5	4	1	16.0	0.2063492063492064	0.6853971971467449	EVT-P-0007565-T02-IM6-1, EVT-P-0012674-T01-IM5-10, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11
167	217	50	4	6	3	2	6.0	0.5238095238095238	0.2808266095756942	EVT-P-0007565-T02-IM6-1, EVT-P-0037180-T01-IM6-7, EVT-P-0039225-T01-IM6-6, EVT-P-0040997-T01-IM6-11

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000756 5-T02-IM6-1	P-000756 5-T02-IM6	P-000756 5	Pancreatic Cancer	PAAD	BAD-RPS6KA4	BAD	in-frame	three_prime	64128886	6	191	True	191	True	exact match	retained	1.0	False												Protein kinase domain; Protein kinase C terminal domain			BAD (NM_032989) - RPS6KA4 (NM_003942) rearrangement: c.379-374:BAD_c.571-55:RPS6KA4 inv | Protein Fusion: in frame {BAD:RPS6KA4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000756 5-T06-IM7-2	P-000756 5-T06-IM7	P-000756 5	Pancreatic Cancer	PAAD	BAD-RPS6KA4	BAD	in-frame	three_prime	64128886	6	191	True	191	True	exact match	retained	1.0	False												Protein kinase domain; Protein kinase C terminal domain			BAD (NM_032989) - RPS6KA4 (NM_003942) rearrangement: c.379-374:BAD_c.571-55:RPS6KA4 inv | Protein Fusion: in frame {BAD:RPS6KA4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003175 6-T01-IM6-4	P-003175 6-T01-IM6	P-003175 6	Non-Small Cell Lung Cancer	LUCA	FERMT3-RPS6KA4	FERMT3	out-of-frame	three_prime	64126817	2	19	True	19	True	exact match	retained	1.0	False												Protein kinase domain; Protein kinase C terminal domain			FERMT3 (NM_031471) - RPS6KA4 (NM_003942) rearrangement: c.1671-205:FERMT3_c.56-33:RPS6KA4del | Protein Fusion: out of frame {FERMT3:RPS6KA4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002231 0-T01-IM6-5	P-002231 0-T01-IM6	P-002231 0	Breast Cancer	IDC	GPR137-RPS6KA4	GPR137	in-frame	five_prime	64137399	14	599	True	599	True	exact match	disrupted	0.713178294573643 5	True	partner_lacks_domain	PF00069					54			False		Protein kinase C terminal domain		Protein kinase domain; Protein kinase domain	RPS6KA4 (NM_003942) - GPR137 (NM_001170726) Rearrangement: c.1797+34:RPS6KA4_c.160-775:GPR137dup | Protein Fusion: in frame {RPS6KA4:GPR137}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003922 5-T01-IM6-6	P-003922 5-T01-IM6	P-003922 5	Breast Cancer	IDC	MARK2-RPS6KA4	MARK2	in-frame	three_prime	64128888	6	191	True	191	True	exact match	retained	1.0	False												Protein kinase domain; Protein kinase C terminal domain			MARK2 (NM_001039469) - RPS6KA4 (NM_003942) rearrangement: c.55-2479 8:MARK2_c.571-53:RPS6KA4del | Protein Fusion: in frame {MARK2:RPS6KA4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003718-0-T01-IM6-7	P-003718-0-T01-IM6	P-003718-0	Colorectal Cancer	READ	PLEKHA7-RPS6KA4	PLEKHA7	unknown	three_prime	64128995	6	209	False	209	True	exact match	retained	1.0	False												Protein kinase domain; Protein kinase C terminal domain			PLEKHA7 (NM_175058) - RPS6KA4 (NM_003942) rearrangement: c.222-55887:PLEKHA7_c.625:RPS6KA4 inv | Protein Fusion: mid-exon { PLEKHA7: RPS6KA4 }	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001267 4-T01-IM5-10	P-001267 4-T01-IM5	P-001267 4	Breast Cancer	IDC	RPS6KA4-DOCK2	DOCK2	in-frame	three_prime	64135589	10	358	True	358	True	exact match	retained	1.0	False													Protein kinase domain; Protein kinase do main		Protein kinase C terminal domain	DOCK2 (NM_004946) - RPS6KA4 (NM_003942) rearrangement: t(5;11)(q35.1;q13.1)(chr5:g.169409522::chr11:g.64135589) | Protein fusion: in frame (DOCK2-RPS6KA4)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004097-T01-IM6-11	P-004097-T01-IM6	P-004097	Germ Cell Tumor	TCCA	RPS6KA4-EEFSEC	EEFSEC	out-of-frame	five_prime	64129088	6	217	True	217	True	exact match	lost	0.0	False														Protein kinase domain; Protein kinase C terminal domain		RPS6KA4 (NM_003942) - EEFSEC (NM_021937) rearrangement: t(3;11)(q21.3;q13.1)(chr3:g.128107142::chr11:g.64129088)| Protein Fusion: out of frame {RPS6KA4::EEFSEC}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002746-T01-IM6-12	P-002746-T01-IM6	P-002746	Esophagogastric Cancer	STAD	RPS6KA4-MACROD1	MACROD1	unknown	five_prime	64138071	16	665	False	664	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.9689922480620154	True	partner_lacks_domain	PF00069					173			False		Protein kinase C terminal domain		Protein kinase domain; Protein kinase domain	RPS6KA4 (NM_003942) - MACROD1 (NM_014067) rearrangement: c.1994:RPS6KA4_c.518-47220:MACROD1inv | Protein Fusion: mid-exon { RPS6KA4:MACROD1}	NA

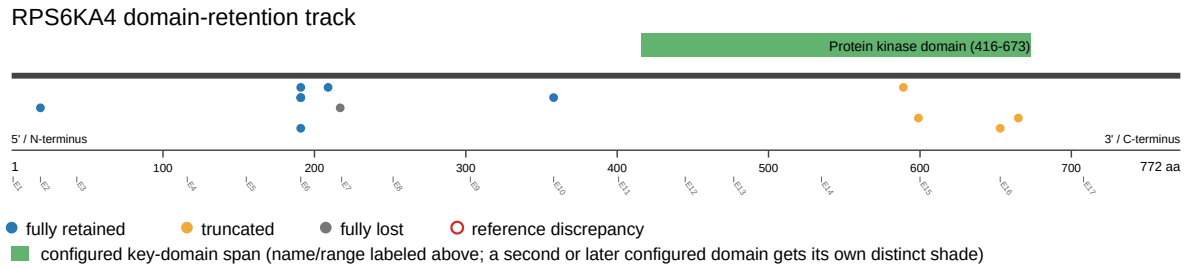
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005007-T01-IM6-20	P-005007-T01-IM6	P-005007	Soft Tissue Sarcoma	MFH	RPS6KA4-SLC22A9	SLC22A9	out-of-frame	five_prime	64137930	15	653	True	652	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.9224806201550387	True	partner_lacks_domain	PF00069					358			False		Protein kinase C terminal domain		Protein kinase domain; Protein kinase domain	RPS6KA4 (NM_003942) - SLC22A9 (NM_080866) rearrangement: c.1957+74: RPS6KA4_c.1073+9725: SLC22A9dup | Protein Fusion: out of frame {RPS6KA4:SLC22A9}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004870 2-T01-IM6 -21	P-004870 2-T01-IM6	P-004870 2	Non-Small Cell Lung Cancer	LUAD	STIP1-RPS6KA4	STIP1	unknown	three_prime	64137333	14	589	False	589	True	exact match	disrupted	0.32945736434108525	True	partner_lacks_domain	PF00069					341			False			Protein kinase C terminal domain	Protein kinase domain; Protein kinase domain	STIP1 (NM_006819) - RPS6KA4 (NM_003942) rearrangement: c.1024-379 :STIP1_c.1765: RPS6KA4 del | Protein Fusion: mid-exon { STIP1:RPS6KA4}	NA

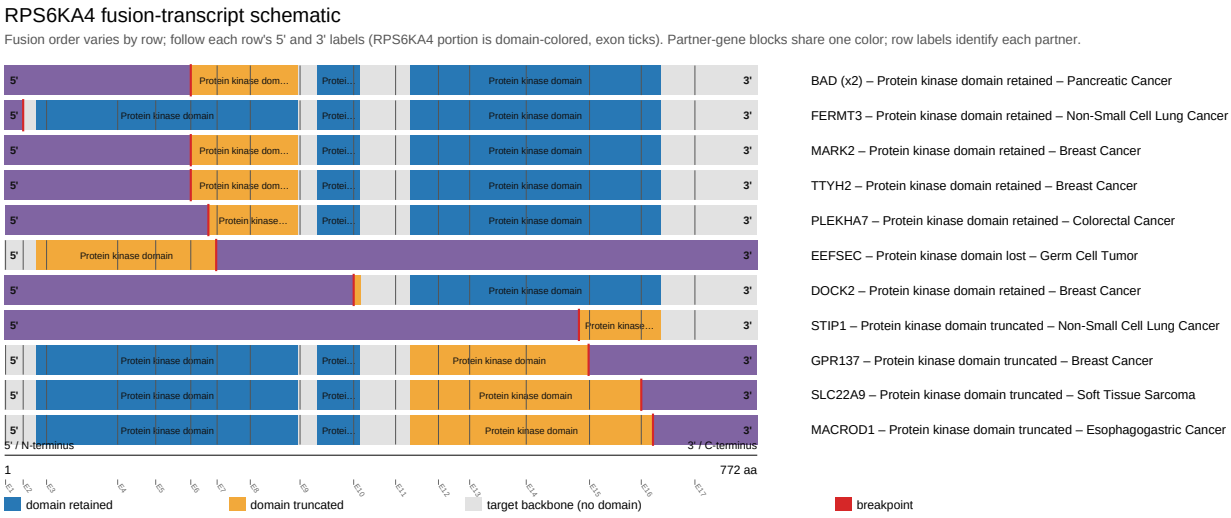
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003922 5-T01-IM6-22	P-003922 5-T01-IM6	P-003922 5	Breast Cancer	IDC	TTYH2-RPS6KA4	TTYH2	unknown	three_prime	64128889	6	191	True	191	True	exact match	retained	1.0	False													Protein kinase domain; Protein kinase C terminal domain			TTYH2 (NM_032646) - RPS6KA4 (NM_003942) rearrangement: t(11;17)(q13.1;q25.1)(chr11:g.64128889::chr17:g.72227047) | Protein Fusion: mid-exon {TTYH2:RPS6KA4}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.